

IoT Cloud Integration & Real-Time Remote Monitoring

Task- 4

Objective:

The objective of Task-4 is to **upgrade your automation system** into a **cloud-connected IoT device** that can send live sensor data to the internet and allow remote monitoring from anywhere.

By completing this task, you will learn:

- ✓ How embedded devices communicate with cloud platforms
- ✓ How to publish live sensor readings online
- ✓ How IoT dashboards are created in real-world projects
- ✓ Essential skill for smart devices industry (Smart Home, Industry 4.0)



Task Description

You will take your **Task-3 Smart Automation System** and connect it to a **free cloud IoT platform** so that sensor readings are visible in **real-time on a web/mobile dashboard**.

✓ Step 1 – Choose any one IoT Cloud Platform (Free)

Platform	Difficulty	Best For
ThingSpeak	Easy	Beginners
Blynk Cloud	Easy + Mobile App	Attractive dashboards
Arduino IoT Cloud	Medium	Arduino users
Adafruit IO	Medium	Data visualization

Recommended: **ThingSpeak (Beginner)** or **Blynk (Mobile App UI)**

✓ Step 2 – Upgrade Hardware/Simulator

If using:

- **ESP32** → Built-in WiFi (recommended)
- **Arduino UNO** → Use WiFi Module ESP8266 (optional) or Wokwi simulator with ESP32

If student doesn't have hardware →

✓ Use **Wokwi.com ESP32 simulator** (cloud compatible)

✅ Step 3 – Send Real-Time Sensor Data to Cloud

What you need to implement:

1. Read sensor values (as in Task-3)
2. Connect to WiFi (ESP32)
3. Send data to IoT cloud platform API
4. Display graph/chart on cloud dashboard

Working Logic (Simple Flow)

arduino

Sensor → Microcontroller → WiFi → IoT Cloud Server → Dashboard → User

✅ Step 4 – Visual Dashboard

Your cloud panel should show:

- **Live reading** (e.g., Temperature / Light / Motion)
- **Graph chart** of changing values (if supported)
- Optional: Trigger alerts

Example Pseudocode (Cloud + Sensor)

bash

🗨 Ask ChatGPT

Connect to WiFi

Loop:

```
read sensor value
upload to cloud channel/variable
wait few seconds
```

(During final submission you will replace this with actual code based on chosen platform)

Free Resources (Videos + Docs)

Topic	Link
ESP32 WiFi Setup	https://www.youtube.com/watch?v=1pwqS_NUG7Q
ThingSpeak with ESP32	https://www.youtube.com/watch?v=Zt9NbtH0U_s
Blynk IoT Setup	https://www.youtube.com/watch?v=PsXF6RjV3LQ
Wokwi ESP32 Cloud Example	https://wokwi.com/projects/305459212104740866

Deliverables

You must submit:

- ✓ Screenshot of IoT dashboard
- ✓ Screenshot of circuit (hardware or simulator)
- ✓ Source code with comments
- ✓ Explanation (How device sends data to cloud)
- ✓ Cloud dashboard link (optional, if sharable)
- ✓ (Optional Bonus) Short recording of live update

✓ Outcome of Task-4

After this task, students will have:

- ✓ A working **cloud-connected IoT system**
- ✓ Experience in **real-time device-to-cloud communication**
- ✓ Knowledge of **IoT dashboard creation**
- ✓ A **portfolio-grade mini IoT product**